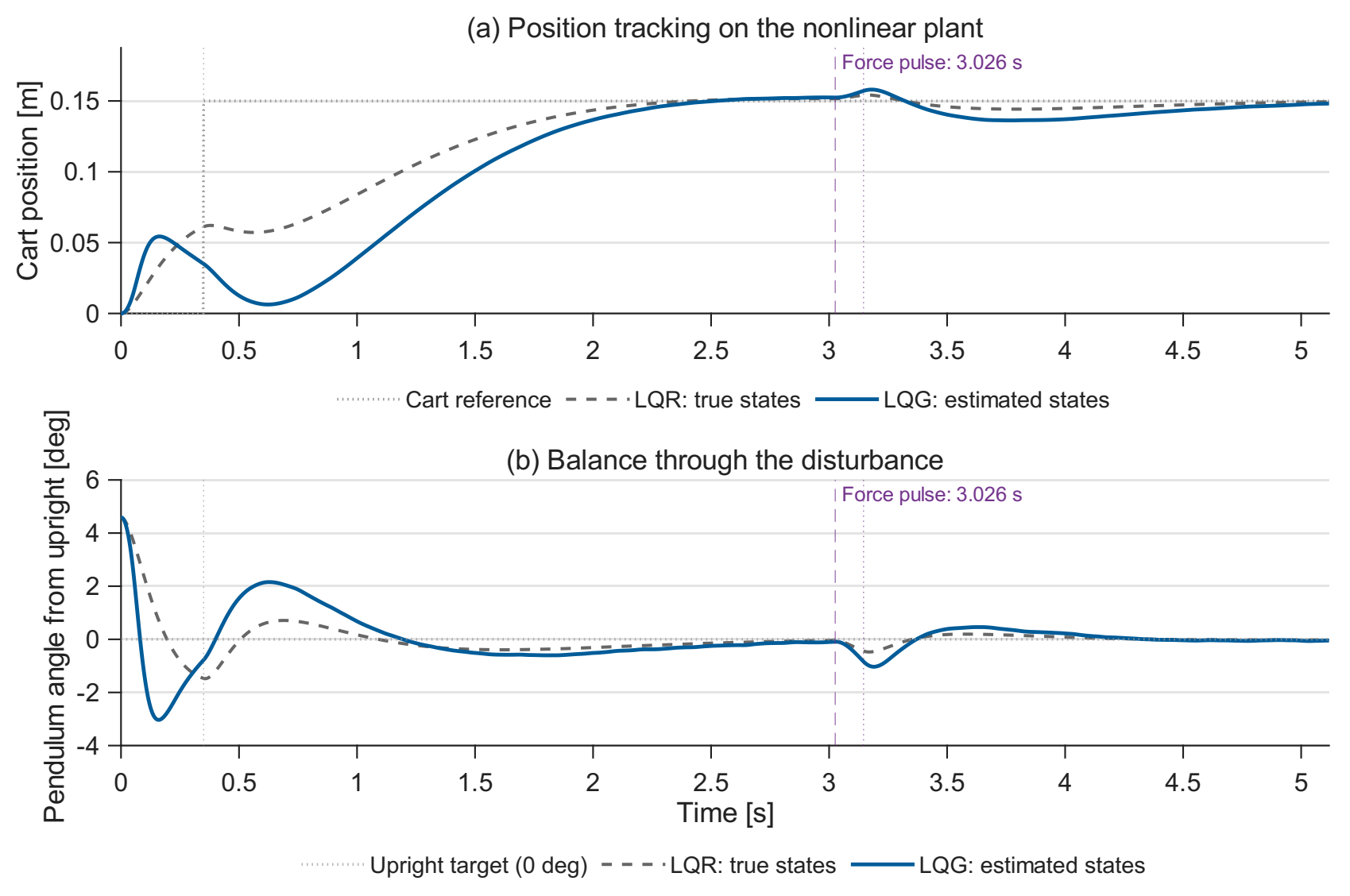

(a) Position tracking on the nonlinear plant
Force pulse: 3.026 s
Cart position [m]
0.15
0.1
0.05
0
0
0.5
1
1.5
2
2.5
3
3.5
4
4.5
5
Cart reference
LQR: true states
LQG: estimated states
(b) Balance through the disturbance
Force pulse: 3.026 s
Pendulum angle from upright [deg]
6
4
2
0
-2
-4
0
0.5
1
1.5
2
2.5
3
3.5
4
4.5
5
Time [s]
Upright target (0 deg)
LQR: true states
LQG: estimated states